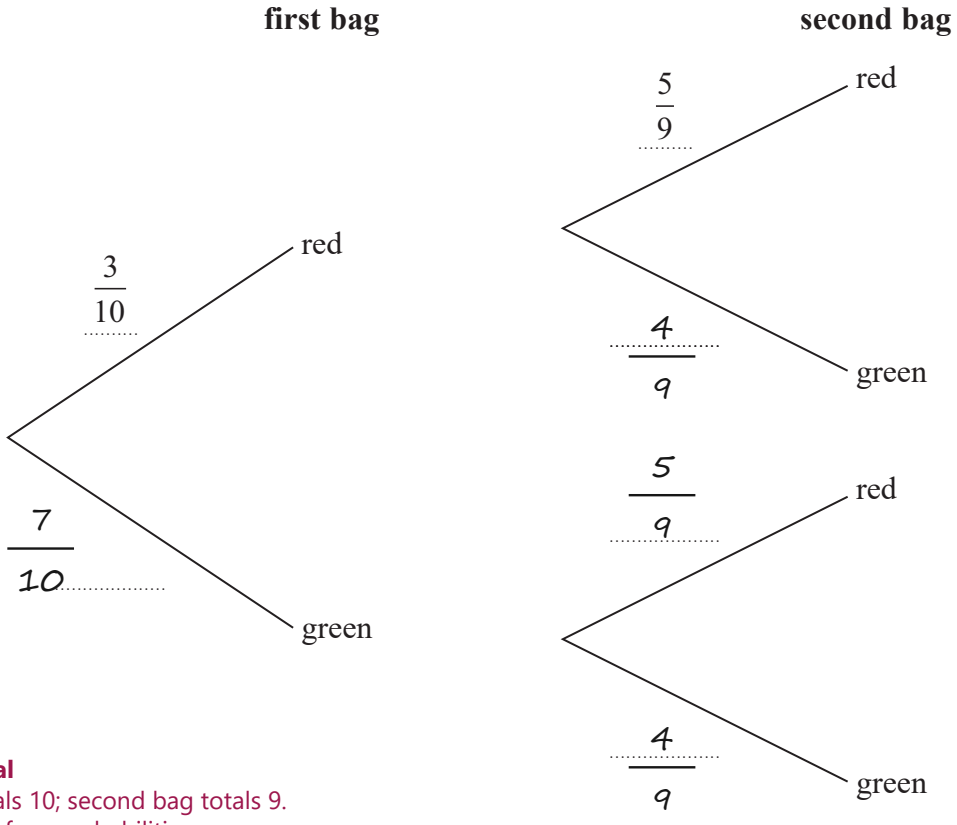
27 Amina has two bags.

In the first bag there are 3 red balls and 7 green balls.
In the second bag there are 5 red balls and 4 green balls.

Amina takes at random a ball from the first bag.
She then takes at random a ball from the second bag.

(a) Complete the probability tree diagram.



2 marks total
First bag totals 10; second bag totals 9.
Complete all four probabilities.

(2)

(b) Work out the probability that Amina takes two red balls.

Two reds means red from the first bag AND red from the second.

$$\frac{3}{10} \times \frac{5}{9} = \frac{15}{90} = \frac{1}{6}$$

1 mark
Multiply along the red-red route

1 mark
Correct probability

$\frac{1}{6}$

(2)

(Total for Question 27 is 4 marks)